EECS 2311 Team 10 Planning Document

Humayun Ejaz
Alireza Hakim
Johnson Han
Manoj Ravichandran
Daniel Rusetski
Umer Shaikh

Original Plan Details

Our team would like to create a location sharing social platform meant for photographers of various levels to share and promote their photos and where they took them.

Vision Statement: FotoFinder

Project Overview

FotoFinder is a community-driven photography location-sharing platform designed to help photographers and hobbyists discover scenic spots for photography. It eliminates the guesswork and time-consuming travel involved in finding the perfect location by allowing users to pin and explore photography-friendly locations contributed by the community.

Target Users

The primary users of FotoFinder are:

- Professional photographers looking for new and unique photography spots.
- Hobbyist photographers who want to explore new places and improve their craft.
- Travelers and outdoor enthusiasts interested in capturing stunning landscapes.

Project Value & Differentiation

- **Convenience:** Users can find pre-vetted locations with high-quality image previews, eliminating the need to scout locations manually.
- **Community Engagement:** Users can rate locations, leave comments, and follow fellow photographers, creating an interactive and supportive environment.
- **Location Accuracy:** Integrates Leaflet Map API and Nominatim Geocoder API to provide precise coordinates and address details for each pinned location.
- **Cloud-Based Storage:** All user data, images, and location pins are securely stored in Firebase, ensuring fast and seamless access across devices.
- **Smart Filtering:** Users can search by categories (urban, nature, night photography, etc.), weather conditions, or user ratings to find the perfect spot.

Technical Implementation

- **Frontend:** Next.js or Vite with TypeScript, using the Leaflet Map API for map visualization.
- **Backend:** Firebase for user authentication, database management (RTDB), and image storage.
- **Testing:** Jest for unit and integration testing to ensure a robust and scalable system.

Success Criteria

FotoFinder will be considered successful if it:

- Attracts active users who pin locations and contribute high-quality images.
- Provides a smooth and intuitive user experience, allowing users to explore and filter locations effortlessly.
- Ensures secure and reliable authentication and data storage using Firebase.
- Encourages community engagement through interactions like comments, ratings, and follows.
- Implements efficient search and filtering capabilities to enhance usability.

By Humayun

Meeting #1 Log Entry

Meeting 1 - 10:01 AM 2025-02-03
The TA suggested to incorporate features that will make our app unique compared to other social media-adjacent apps, so we should incorporate things that are unique to photographers (e.g. camera settings).

Potential unique features:

- Posts including the camera settings/specs that were used to take the picture
- Review system so people can rate recommended locations and users (kind of like Google Maps or Uber)
- User profile can include current camera and maybe a camera history (history of cameras the user has had)
- General recommendations/tips for camera settings
- Posts can include the raw file and the edited picture

General features:

- Dark mode
- Follower/following system

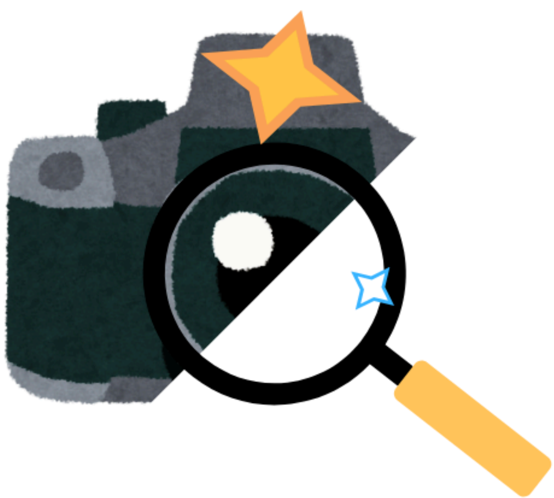
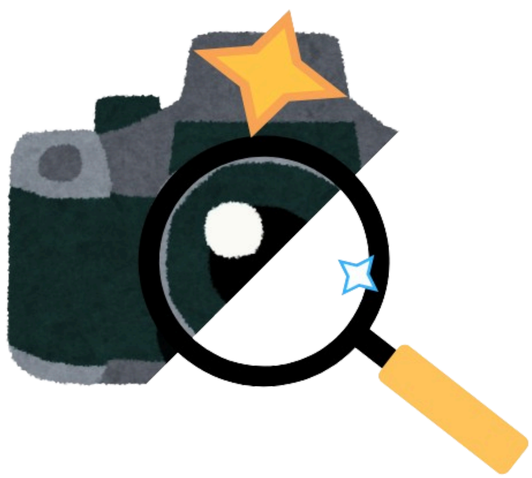
Initial Suggested Delegation of Tasks

- Humayun Ejaz:
  - Vision statement
- Alireza Hakim:
  - User stories
  - Big user stories
  - Creating Planning Document
- Johnson Han:
  - User stories
  - General code development
- Manoj Ravichandran:
  - General development
- Daniel Rusetski:
  - Video interview
  - Code environment setup
  - GitHub setup
- Umer Shaikh:
  - Frontend development

Application Logo

Initial Logo Design:

[Canva Link]

By Alireza

Technical Features to Add Based on Detailed User Stories

User Story Number	Technical Details	Deadline	Team Members
1			

Coding Plans

1. Create directories for reusable components and pages. Don’t need to add reactivity yet, but should be a single page application.
2. Create the canvas for the home page with HTML and Tailwind CSS. Essentially just creating divs that match the layout of the concept screenshots sent. No styling necessary yet, but can be started.
3. Set up nominatim geocoder API and leaflet map after canvas is finished.
4. Set up backend (Firebase or AWS) and ensure a database can be queried and the frontend interfaces the backend.
5. Set up testing framework and run tests with non-persistent data.

By Johnson

Meeting 2 - 9:49 AM 2025-02-24

ITR0 and ITR1 are done, just need to continue coding, especially our code foundation. All our planning documents are done and uploaded to the GitHub. Currently making plans to meet and catch up on any shortcomings in terms of the project's code.

Meeting 3 - 8:02 PM 2025-02-24

Coding todo list:

- Setup Map API/Basic UI (Leaflet)
- Setup FireBase
- Create and Connect Backend to Frontend
- Add posts to test pins throughout the map
- Add a separate Post feature
- Add user profiles and authentication through AWS
- Figma visuals for how we want the website to look like

Action Items:
- Create concepts of visuals on Figma
- Get familiar with relevant tools that we will use (codepen.io)